

Determinant Mass on the Actual Collision Core at a Fixed Shift: Forced Matchings, Alternating Holonomy, and Cauchy–Binet Lower Frames

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Abstract

The preceding fixed-shift analysis identifies an actual unit-diagonal residual Gram and proves subpower reduced-cell codegree for every fixed row pair, but does not control its global smallest eigenvalue. We open a determinant route on the literal full-output matrix. For every square full-column minor, repeated degree-one peeling factors off exactly all pivots found by the iterative degree-one rule. A residual core without a perfect matching has zero determinant. If the core admits a perfect matching, normalizing that matching to the identity makes every other determinant term a vertex-disjoint family of directed alternating cycles; diagonal gauge leaves only the cycle-holonomy parameters. If $\mathcal{A}_\phi = \text{per}(I + |E|) - 1$ is the total absolute activity of the nonempty cycle families, P_{for} is the product of the forced-pivot magnitudes, and P_ϕ is the core matching-pivot product, then

$$|\det T| \geq P_{\text{for}} P_\phi (1 - \mathcal{A}_\phi)_+.$$

We next aggregate rather than select minors. For a synthesis matrix B with r unit columns and Gram $K = B^*B$, Cauchy–Binet and the trace constraint give, for $r \geq 2$, the sharp conversion

$$\mathfrak{D}(B) := \sum_{|Y_0|=r} |\det B_{Y_0}|^2 = \det K, \quad \lambda_{\min}(K) \geq \left(\frac{r-1}{r}\right)^{r-1} \mathfrak{D}(B) \geq e^{-1} \mathfrak{D}(B).$$

The case $r = 1$ is exact with constant one. Thus certified determinant mass transfers to the physical Gram without the dimension-exponential norm penalty of a one-minor estimate. The theorem is sharp, but the route is not automatically sufficient: a path-Gram family has a uniform spectral gap while its determinant decays exponentially. The remaining arithmetic task for this certificate is to lower-bound the determinant mass of every actual full-support collision component, or the corresponding certified-minor mass; a global determinant-mass lower bound is a stronger sufficient condition. A direct spectral or signed-reassembly argument remains an independent route. No such physical lower bound, parity improvement, or prime-pair conclusion is claimed.

Keywords: fixed shift; determinant mass; collision core; forced matching; alternating cycle; holonomy; Cauchy–Binet; lower frame.

Convention. Arithmetic specializations use one fixed nonzero shift h_0 , literal full output coordinates (γ, s, j) , actual nonzero coefficients, all active physical rows, and raw counting-measure atomic norms. The shift is not specialized to 2.

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1 Introduction

TPC-65 reduced the represented-reference gate to the least eigenvalue of an actual activated-range Gram [1]. TPC-66 then opened its literal fixed-shift geometry: after retaining every active physical row and every full output key (γ, s, j) , the normalized row columns have norm one; fixed row pairs meet in only $X^{o(1)}$ reduced cells on the critical cofactor block [2]. That is genuine pair thinning, but not a lower-frame theorem. The full γ -decorated support, right degrees, energy concentration, and global cycles can still be large.

This paper asks the smallest next question:

Can the global lower frame be certified by the determinant mass of the actual full-output matrix, and exactly which collision geometry controls that mass?

The answer has a positive exact part and a sharp negative boundary.

First, a square physical minor is reduced to its nonpeelable collision core. Every degree-one support vertex forces one matching pivot and one Laplace factor. Once a perfect matching of the core is placed on the diagonal, the normalized matrix is $I + E$. The determinant expansion is then not an unstructured sum: every nonidentity term is exactly a vertex-disjoint family of directed alternating cycles. A diagonal similarity removes all phases on a spanning forest, leaving only chord holonomies. This generalizes the unique-matching and one-cycle examples of TPC-66 without replacing the literal coefficients by generic fillings.

Second, the absolute activity

$$\mathcal{A}_\phi = \text{per}(I + |E|) - 1$$

gives a sharp dominant-matching certificate. If $\mathcal{A}_\phi < 1$, the chosen matching term cannot be canceled by all cycle families combined. A useful sufficient version sums simple directed cycle weights. Its $\log 2$ threshold is deliberately presented only as a consequence of an exponential majorant, not as a sharp physical threshold.

Third, we avoid the main quantitative defect of a one-minor argument. If B has r unit columns and $K = B^*B$, all full-column minors satisfy

$$\sum_{|Y_0|=r} |\det B_{Y_0}|^2 = \det K.$$

Since $\text{tr } K = r$, the other $r - 1$ eigenvalues can consume only the remaining trace. For $r \geq 2$ this yields

$$\lambda_{\min}(K) \geq c_r \det K, \quad c_r = \left(\frac{r-1}{r}\right)^{r-1} \geq e^{-1}.$$

while $c_1 = 1$. The constant is sharp. Hence a fixed-power lower bound for the aggregate determinant mass transfers with zero exponent loss, unlike the $\|T\|^{-2(r-1)}$ conversion attached to one arbitrary minor.

The limitation is equally important. Determinant mass multiplies all eigenvalues. It may therefore be exponentially small in the number of rows even when the least eigenvalue stays uniformly positive. We exhibit this behavior on a path Gram. Failure of the determinant route is consequently not a physical spectral obstruction; it is a certified reason to return to the direct comparison/forest route of TPC-66 or to open the later signed reassembly door.

Claim levels

The peeling, alternating-cycle expansion, holonomy count, activity bounds, Cauchy–Binet identity, trace conversion, and countermodels are L0. Identifying their matrix with the literal fixed- h_0 full-output synthesis and retaining all active rows is L1. A polynomial lower bound for the actual determinant mass or physical Gram gap would be L2 and is not proved. Reduced cells are not substituted for (γ, s, j) , no random phase or generic filling is used, and no property special to $h_0 = 2$ enters.

2 The literal full-output minor interface

TPC-65 identifies the represented lower-frame constant with the smallest eigenvalue of the actual activated row Gram, and TPC-66 identifies the literal row columns and their support geometry [1, 2]. We now fix the matrix on which every determinant in this paper is computed. The output coordinate remains the complete physical key (γ, s, j) ; no reduced-cell quotient is made in the matrix definition.

2.1 All active rows and all full outputs

Fix X , one nonzero shift h_0 , one declared source system, and all external block, direct-residue, orbit, mask, profile, and endpoint data before choosing a row vector. Let $\mathcal{U}_X$ be the actual nonzero terminal atoms in this fixed scope. For $u \in \mathcal{U}_X$, write

$$a_u := t_{h_0}(u)c_u \neq 0$$

for its literal coefficient, and retain its declared maps $m(u)$, $r(u)$, $j(u)$, and $\gamma(u)$, together with the inherited row factor $d_{m(u)}$. Define

$$T_m := \sum_{\substack{u \in \mathcal{U}_X \\ m(u)=m}} |a_u|^2.$$

Let

$$\mathcal{M}_X^{\text{act}} := \{m : T_m > 0\}$$

be the complete set of terminal-active physical rows. For a full output

$$y = (\gamma, s, j),$$

write the literal coherently summed row coefficient as

$$a_{y,m} := \sum_{\substack{u \in \mathcal{U}_X: q(u)=y \\ m(u)=m}} a_u, \quad q(u) = (\gamma(u), d_{m(u)}r(u), j(u)). \quad (1)$$

Every summand and every zero in (1) belongs to the fixed physical carrier. On the declared source system, the row-singleton theorem of TPC-66 says that the displayed sum is either empty or contains one literal terminal atom. In particular,

$$T_m = \sum_y |a_{y,m}|^2 \quad (m \in \mathcal{M}_X^{\text{act}}). \quad (2)$$

Define the complete actual output set

$$\mathcal{Y}_X^{\text{full}} := \{(\gamma, s, j) : a_{(\gamma,s,j),m} \neq 0 \text{ for at least one } m \in \mathcal{M}_X^{\text{act}}\}. \quad (3)$$

It contains every nonzero full output coordinate generated by every active row. Put

$$r_X := |\mathcal{M}_X^{\text{act}}|.$$

If $r_X = 0$, the represented gate is empty and no lower-frame or determinant constant is assigned. Henceforth assume $r_X \geq 1$.

Definition 2.1 (Literal normalized full-output matrix). The matrix

$$B_X : \ell^2(\mathcal{M}_X^{\text{act}}) \longrightarrow \ell^2(\mathcal{Y}_X^{\text{full}}) \quad (4)$$

has entries

$$(B_X)_{y,m} := \frac{a_{y,m}}{\sqrt{T_m}}. \quad (5)$$

Its actual normalized Gram and lower-frame constant are

$$K_X := B_X^* B_X, \quad \beta_X := \lambda_{\min}(K_X) = \sigma_{\min}(B_X)^2. \quad (6)$$

Every column of B_X has norm one:

$$\sum_{y \in \mathcal{Y}_X^{\text{full}}} |(B_X)_{y,m}|^2 = 1. \quad (7)$$

If z is an original row coefficient vector and $x_m = \sqrt{T_m} z_m$, then the represented quotient relative to terminal energy is

$$\frac{\|B_X x\|^2}{\|x\|^2}.$$

Thus β_X is exactly the TPC-65 activated-range constant on all active rows, not the lower constant of a selected collision-free packet.

2.2 Full minors and determinant mass

Fix arbitrary orderings of $\mathcal{M}_X^{\text{act}}$ and $\mathcal{Y}_X^{\text{full}}$. For $I \subseteq \mathcal{Y}_X^{\text{full}}$ with $|I| = r_X$, let

$$B_X[I] := ((B_X)_{y,m})_{y \in I, m \in \mathcal{M}_X^{\text{act}}} \quad (8)$$

be the square row restriction using every active physical row. Define

$$\mathfrak{d}_X(I) := |\det B_X[I]|^2, \quad (9)$$

$$\mathfrak{D}_X := \sum_{\substack{I \subseteq \mathcal{Y}_X^{\text{full}} \\ |I| = r_X}} \mathfrak{d}_X(I). \quad (10)$$

The sum is zero when $|\mathcal{Y}_X^{\text{full}}| < r_X$. The adjective *full* means that a minor uses all active columns and literal full-output rows. It does not mean that every output coordinate is retained in one square minor. The sum in (10), rather than an adaptively chosen minor, is the canonical determinant mass.

Proposition 2.2 (Actual rank criterion). *The following are equivalent:*

- (i) $\mathfrak{D}_X > 0$;
- (ii) *at least one literal full minor $B_X[I]$ is nonsingular;*
- (iii) *B_X has full column rank;*
- (iv) $\ker B_X = \{0\}$;
- (v) *K_X is positive definite, equivalently $\beta_X > 0$.*

Proof. The determinant mass is a finite sum of nonnegative numbers, so it is positive exactly when one full minor is nonzero. The ordinary rank-by-minors criterion gives the equivalence with full column rank. Finally, $\ker(B_X^* B_X) = \ker B_X$, which proves the last two equivalences. No quantitative Cauchy–Binet lower inequality is used here. $\square$

The later determinant-mass section will relate $\mathfrak{D}_X$ quantitatively to the lower spectrum. We do not anticipate or reprove that inequality in the interface.

2.3 Support Hall defect versus the literal rank defect

Let $\mathcal{H}_X^{\text{full}}$ be the bipartite support graph of B_X . Its left vertices are $\mathcal{M}_X^{\text{act}}$, its right vertices are $\mathcal{Y}_X^{\text{full}}$, and

$$m \sim y \iff (B_X)_{y,m} \neq 0.$$

For $S \subseteq \mathcal{M}_X^{\text{act}}$, let

$$N_X^{\text{full}}(S) := \{y : \text{there is } m \in S \text{ with } m \sim y\},$$

and write

$$B_{X,S} := B_X[N_X^{\text{full}}(S), S] : \ell^2(S) \longrightarrow \ell^2(N_X^{\text{full}}(S)).$$

Define the support Hall defect and the literal rank defect:

$$\mathfrak{h}_X^{\text{supp}} := \max_{S \subseteq \mathcal{M}_X^{\text{act}}} (|S| - |N_X^{\text{full}}(S)|)_+, \quad (11)$$

$$\mathfrak{h}_X^{\text{act}} := \max_{S \subseteq \mathcal{M}_X^{\text{act}}} (|S| - \text{rank } B_{X,S}). \quad (12)$$

Here $x_+ := \max\{x, 0\}$. The first depends only on the full support. The second uses the literal complex entries and is therefore an actual, rather than generic, rank defect; despite the inherited symbol $\mathfrak{h}_X^{\text{act}}$, it is not called a Hall defect.

Theorem 2.3 (Literal rank-defect–kernel equivalence). *One has*

$$\boxed{\mathfrak{h}_X^{\text{supp}} \leq \mathfrak{h}_X^{\text{act}} = \dim \ker B_X = \dim \ker K_X.} \quad (13)$$

Consequently the following are equivalent:

- (i) $\ker B_X = \{0\}$;
- (ii) for every $S \subseteq \mathcal{M}_X^{\text{act}}$, $\text{rank } B_{X,S} = |S|$;
- (iii) $\mathfrak{h}_X^{\text{act}} = 0$;
- (iv) $\mathfrak{D}_X > 0$.

Ordinary Hall inequalities

$$|N_X^{\text{full}}(S)| \geq |S| \quad (S \subseteq \mathcal{M}_X^{\text{act}}) \quad (14)$$

are necessary for these conditions but are not sufficient for the one literal coefficient array.

Proof. Since $\text{rank } B_{X,S} \leq |N_X^{\text{full}}(S)|$, the first inequality in (13) follows. The nullity of $B_{X,S}$ is

$$|S| - \text{rank } B_{X,S} = \dim\{x \in \ker B_X : \text{supp } x \subseteq S\},$$

and is therefore at most $\dim \ker B_X$. Taking $S = \mathcal{M}_X^{\text{act}}$ gives equality, proving the middle identity. The Gram has the same kernel as B_X . The equivalences now follow from [proposition 2.2](#). A violation of (14) forces a positive support defect and hence a kernel. Conversely, cycle cancellation can give an actual kernel even when all cardinal Hall inequalities hold, as already demonstrated at L0 in TPC–66. $\square$

2.4 Why reduced-cell codegree is auxiliary

Let

$$\pi : \mathcal{Y}_X^{\text{full}} \longrightarrow \mathcal{Y}_X^{\text{red}}, \quad \pi(\gamma, s, j) = (s, j). \quad (15)$$

The TPC-66 arithmetic codegree bounds the number of reduced cells (s, j) in which a fixed pair of rows can meet. This is useful for locating possible nonzero Gram summands, but π can identify many orthogonal γ -coordinates. In particular:

- (a) a square matrix indexed by reduced cells is not a row restriction of B_X ;
- (b) a matching in the projection of the *actual* full-support graph to distinct reduced cells has a support lift: choose one nonzero full coordinate over each matched incidence, and distinct reduced cells force the chosen full coordinates to be distinct. This lift gives a support matching, not a numerical determinant lower bound, because other permutation terms may cancel it. By contrast, a matching in the larger arithmetic supergraph used for the TPC-66 codegree count need not have any literal nonzero lift;
- (c) reduced Hall deficiency need not imply full Hall deficiency, because one reduced cell can have many full-output lifts; and
- (d) reduced pair codegree does not determine $\mathfrak{D}_X$, $\ker B_X$, or β_X .

All full minors, neighborhoods, ranks, and kernels in this paper are therefore computed before applying π . Reduced-cell codegree may prune the list of possible full collisions; it never replaces their literal γ -multiplicity or coefficient values.

2.5 L1 quantifier ledger

The arithmetic interface has the following fixed order of quantifiers.

- (L1.1) Fix $h_0 \neq 0$, X , the source system, all multiscale/direct cells, residue and orbit classes, masks, profiles, and literal terminal support.
- (L1.2) Form the actual nonzero entries (1), then retain *every* row with $T_m > 0$ and every full output touched by those rows.
- (L1.3) Normalize columns only by their physical terminal masses T_m . No fiber degree, row multiplicity, or reduced-cell multiplicity is absorbed into the norm.
- (L1.4) Construct the full support graph, all full minors, determinant mass, literal ranks and nullity, and support components from this fixed matrix, before testing any coefficient vector.
- (L1.5) Evaluate a lower-frame statement uniformly on the complete active row space. A row with an inconvenient collision pattern is not deleted, and a favorable minor or component is not substituted for the full physical map.

These are exact finite- X definitions and L0 matrix equivalences, specialized at L1 to the literal fixed- h_0 carrier. They prove no positive exposure lower bound, no polynomial lower bound for $\mathfrak{D}_X$ or β_X , no parity estimate, and no prime-pair conclusion.

3 Forced peeling and the actual alternating-cycle core

This section is finite-dimensional. Let

$$T = (t_{y,m})_{y \in \mathcal{Y}_0, m \in \mathcal{M}}$$

be a square matrix, where $|\mathcal{Y}_0| = |\mathcal{M}| = r$. In the arithmetic application T is a full-column minor of the literal physical synthesis matrix: its rows are actual full output coordinates and its columns are all active physical rows. None of the arguments below replaces those coefficients by generic fillings.

The bipartite support graph $\mathcal{H}(T)$ has parts $\mathcal{M}$ and $\mathcal{Y}_0$, with $m \sim y$ precisely when $t_{y,m} \neq 0$. We use the convention that the determinant of the empty matrix is one.

3.1 Exact two-sided degree-one peeling

A degree-one vertex on either side of $\mathcal{H}(T)$ forces its unique incident edge in every nonzero determinant monomial. Iterating this observation gives an exact factorization, not merely a support reduction.

Theorem 3.1 (Two-sided forced-peeling factorization). *Choose any iterative procedure with the following rule. Whenever the current bipartite support graph has a degree-one vertex, delete that vertex, its unique neighbor, and their incident edge. Let $\mathcal{P}_{\text{for}}$ be the ordered set of deleted edges, and let T_{core} be the square matrix left after the deletions. Then there is a sign $\varepsilon_{\text{for}} \in \{-1, 1\}$ such that*

$$\det T = \varepsilon_{\text{for}} \left(\prod_{e \in \mathcal{P}_{\text{for}}} t_e \right) \det T_{\text{core}}. \quad (16)$$

If a zero-degree vertex occurs at any stage, then $\det T = 0$. Otherwise the final core is empty or every one of its bipartite vertices has degree at least two.

Proof. Suppose first that a column vertex m has the unique neighbor y . Laplace expansion down column m gives

$$\det T = (-1)^{\iota(y) + \iota(m)} t_{y,m} \det T_{\widehat{y}, \widehat{m}},$$

where the hats mean that the indicated row and column have been removed. The same formula follows by expansion across a row when the degree-one vertex is on the output side. Iterating these identities proves (16); all Laplace signs are absorbed into ε_{for} .

A zero row or zero column makes the current determinant zero and hence, through the preceding exact expansions, makes $\det T = 0$. If no such vertex occurs, termination without another degree-one vertex leaves either the empty matrix or minimum degree at least two on both sides. $\square$

Remark 3.2 (Order and scope). The factorization is valid for every chosen peeling order; no uniqueness of the residual core is needed. If the support admits a perfect matching, every peeled edge belongs to every perfect matching of the current graph, and the residual support again admits a perfect matching. If no perfect matching remains, its determinant is zero. Thus order dependence cannot turn a singular minor into a certified nonsingular one.

3.2 Normalization by one core matching

Assume from now on that T_{core} is nonempty and that its support admits a perfect matching

$$\phi : V \longrightarrow \mathcal{Y}_{\text{core}},$$

where V indexes the core columns. Reorder the core output rows so that row i is $\phi(i)$, and put

$$p_i := t_{\phi(i),i} \neq 0, \quad D_\phi := \text{diag}(p_i)_{i \in V}. \quad (17)$$

The matching-normalized matrix is

$$N_\phi := (t_{\phi(i),j})_{i,j \in V} D_\phi^{-1} = I + E_\phi, \quad (18)$$

where

$$(E_\phi)_{i,i} = 0, \quad (E_\phi)_{i,j} = \frac{t_{\phi(i),j}}{p_j} \quad (i \neq j). \quad (19)$$

Consequently,

$$|\det T| = \left(\prod_{e \in \mathcal{P}_{\text{for}}} |t_e| \right) \left(\prod_{i \in V} |p_i| \right) |\det(I + E_\phi)|. \quad (20)$$

The ratios in (19), rather than the unnormalized entry magnitudes alone, are the numerical data that control the remaining cancellation.

3.3 Exact alternating-cycle expansion

Let $\mathcal{D}_\phi$ be the directed graph on V having the arc $i \rightarrow j$ exactly when $i \neq j$ and $(E_\phi)_{i,j} \neq 0$. The arc $i \rightarrow j$ represents the off-matching bipartite edge joining row $\phi(i)$ to column j . A directed cycle

$$C = (i_1 \rightarrow i_2 \rightarrow \cdots \rightarrow i_k \rightarrow i_1)$$

therefore lifts to a bipartite cycle alternating, for each ν , the off-matching and matching edges

$$\phi(i_\nu) - i_{\nu+1} \quad \text{and} \quad \phi(i_\nu) - i_\nu.$$

Define its normalized complex weight by

$$w_\phi(C) := \prod_{\nu=1}^k (E_\phi)_{i_\nu, i_{\nu+1}}, \quad i_{k+1} := i_1. \quad (21)$$

Let $\mathfrak{F}_\phi$ be the set of all nonempty families $\mathcal{F}$ of pairwise vertex-disjoint simple directed cycles in $\mathcal{D}_\phi$, and put

$$w_\phi(\mathcal{F}) := \prod_{C \in \mathcal{F}} w_\phi(C), \quad |V(\mathcal{F})| := \sum_{C \in \mathcal{F}} |C|.$$

Theorem 3.3 (Determinant terms are disjoint alternating-cycle families). *There is a weight-preserving bijection between the nonidentity nonzero permutation terms in $\det(I + E_\phi)$ and the families in $\mathfrak{F}_\phi$. Explicitly,*

$$\boxed{\det(I + E_\phi) = 1 + \sum_{\mathcal{F} \in \mathfrak{F}_\phi} (-1)^{|V(\mathcal{F})| - |\mathcal{F}|} w_\phi(\mathcal{F})}. \quad (22)$$

Proof. Expand the determinant over permutations of V . Because the diagonal of E_ϕ is zero, every fixed point of a permutation contributes the diagonal entry one, while every moved vertex i contributes the arc $i \rightarrow \sigma(i)$. The restriction of a permutation to its moved vertices is uniquely a family of pairwise vertex-disjoint cycles of length at least two. Conversely, such a family, extended by the identity off its vertices, is a unique permutation with a nonzero term.

A cycle of length k has permutation sign $(-1)^{k-1}$. Multiplication over the cycles gives the sign

$$(-1)^{\sum_{C \in \mathcal{F}} (|C|-1)} = (-1)^{|V(\mathcal{F})| - |\mathcal{F}|},$$

and its matrix-entry product is $w_\phi(\mathcal{F})$. This proves both the bijection and (22). $\square$

Corollary 3.4 (Acyclic normalized core). *If $\mathcal{D}_\phi$ has no directed cycle, then*

$$\det(I + E_\phi) = 1$$

and the chosen matching is the only perfect matching of the core support. In particular, all possible determinant cancellation is confined to the directed alternating-cycle families.

Proof. The sum in (22) is empty. The same permutation–cycle correspondence shows that a second perfect matching would produce a nonempty directed cycle family. $\square$

3.4 Diagonal similarity and the holonomy dimension

For a diagonal unitary matrix

$$S = \text{diag}(e^{i\theta_i})_{i \in V},$$

the transformation

$$I + E_\phi \mapsto S^{-1}(I + E_\phi)S = I + S^{-1}E_\phi S \quad (23)$$

preserves the determinant and every entry magnitude. It changes an arc phase by

$$\arg(E_{i,j}) \mapsto \arg(E_{i,j}) - \theta_i + \theta_j,$$

while every directed-cycle product (21) is unchanged.

Write

$$\mathcal{E}_{\text{off}} := \{(i, j) : i \neq j, (E_\phi)_{i,j} \neq 0\}, \quad q_\phi := |\mathcal{E}_{\text{off}}|,$$

and let c_ϕ be the number of weakly connected components of $\mathcal{D}_\phi$, isolated vertices included.

Proposition 3.5 (Gauge reduction to chord phases). *There is a diagonal unitary similarity for which the weights on a spanning forest of the underlying weak graph are positive real. Hence, after entry magnitudes are fixed, at most*

$$\boxed{q_\phi - |V| + c_\phi} \quad (24)$$

arc phases remain. All phase dependence of the determinant factors through these residual chord phases, and every directed alternating-cycle weight is a gauge-invariant holonomy.

Proof. Choose one root in each weak component and a spanning forest containing $|V| - c_\phi$ directed arcs after their orientations are forgotten. Starting at the roots, choose the vertex phases recursively so that the selected arc joining each new vertex to the preceding tree is positive real. A forest imposes no inconsistent phase equation. The selected $|V| - c_\phi$ phases have thereby been removed from the q_ϕ off-diagonal phases, leaving at most $q_\phi - |V| + c_\phi$. Diagonal similarity preserves the determinant, and the vertex phases telescope around every directed cycle, proving the last assertion. $\square$

Remark 3.6 (L0 status and nonduplication with TPC–66). The peeling factorization, matching normalization, exact cycle-family expansion, and gauge count are L0 matrix theorems. TPC–66 proved a support-rank hierarchy, an exact formula on forest Gram graphs, and a single uniform-cycle obstruction [2]. The present result does not repeat those statements: it factors an arbitrary square minor down to its nonpeelable matching core and describes *all* of that core’s determinant monomials by disjoint alternating-cycle families.

An L1 application must use an actual full-output minor, all active physical rows, its literal nonzero pivots, and its literal coefficient ratios. Nothing here proves that such a minor exists, that its forced-pivot product is polynomially large, or that its residual holonomies are favorable.

4 Cycle activity and phase-uniform determinant certificates

Retain the matching-normalized core $I + E_\phi$ and the directed graph $\mathcal{D}_\phi$ from [section 3](#). Absolute values of matrices in this section are entrywise. The point is to separate two questions:

- (i) the exact total mass of all nonidentity determinant terms; and
- (ii) coefficient-blind upper bounds for that mass.

The first has the sharp threshold one. A convenient bound through the sum of individual cycle weights has the smaller sufficient threshold $\log 2$, which is not sharp.

4.1 The exact permanent activity

Definition 4.1 (Matching activity). The phase-blind activity of the matching ϕ is

$$\mathcal{A}_\phi := \text{per}(I + |E_\phi|) - 1. \quad (25)$$

Proposition 4.2 (Exact activity identity). *The activity is exactly the total absolute weight of all nonempty vertex-disjoint directed-cycle families:*

$$\mathcal{A}_\phi = \sum_{\mathcal{F} \in \mathfrak{F}_\phi} |w_\phi(\mathcal{F})|. \quad (26)$$

Proof. Expand the permanent of $I + |E_\phi|$. Its identity term is one. The correspondence in [theorem 3.3](#) applies again, now without permutation signs. The term attached to $\mathcal{F}$ is precisely

$$\prod_{C \in \mathcal{F}} \prod_{(i,j) \in C} |(E_\phi)_{i,j}| = |w_\phi(\mathcal{F})|.$$

Subtracting the identity term proves (26). □

Theorem 4.3 (Sharp phase-uniform activity certificate). *Let*

$$P_{\text{for}} := \prod_{e \in \mathcal{P}_{\text{for}}} |t_e|, \quad P_\phi := \prod_{i \in V} |p_i|.$$

Then

$$|\det T| \geq P_{\text{for}} P_\phi (1 - \mathcal{A}_\phi)_+. \quad (27)$$

In particular, $\mathcal{A}_\phi < 1$ implies that the literal minor is nonsingular, uniformly in all phases having the prescribed entry magnitudes.

Proof. By (22), the identity term of $\det(I + E_\phi)$ is one and the sum of the absolute values of all other terms is $\mathcal{A}_\phi$. The reverse triangle inequality gives

$$|\det(I + E_\phi)| \geq (1 - \mathcal{A}_\phi)_+.$$

Multiplication by the exact pivot factors in (20) proves the result. □

Proposition 4.4 (Why the threshold one is sharp). *Neither the threshold 1 nor the coefficient in $(1 - \mathcal{A}_\phi)_+$ can be improved using only the total activity. Indeed, for*

$$I + E = \begin{pmatrix} 1 & x \\ y & 1 \end{pmatrix} \quad (28)$$

one has

$$\mathcal{A}_\phi = |xy|, \quad \det(I + E) = 1 - xy.$$

Taking $x, y > 0$ gives equality in (27); at $xy = 1$, the determinant vanishes and $\mathcal{A}_\phi = 1$. On the other hand, $\mathcal{A}_\phi \geq 1$ does not imply singularity: taking $xy \leq -1$ gives $|\det(I + E)| = 1 + \mathcal{A}_\phi$.

4.2 From single-cycle mass to family activity

Let $\mathcal{C}_\phi$ be the set of simple directed cycles of $\mathcal{D}_\phi$, and define the single-cycle mass

$$\Xi_\phi := \sum_{C \in \mathcal{C}_\phi} |w_\phi(C)|. \quad (29)$$

Theorem 4.5 (Exponential domination of disjoint-cycle activity). *One has*

$$\boxed{\mathcal{A}_\phi \leq \exp(\Xi_\phi) - 1.} \quad (30)$$

Consequently, if

$$\Xi_\phi < \log 2, \quad (31)$$

then

$$\boxed{|\det T| \geq P_{\text{for}} P_\phi (2 - e^{\Xi_\phi}) > 0.} \quad (32)$$

Proof. Expand the exponential:

$$e^{\Xi_\phi} = \sum_{k=0}^{\infty} \frac{1}{k!} \left(\sum_{C \in \mathcal{C}_\phi} |w_\phi(C)| \right)^k.$$

For a family of k distinct vertex-disjoint cycles, its weight appears exactly $k!$ times in the k -th power, once for every ordering of its cycles, and hence with coefficient one after division by $k!$. The exponential also contains nonnegative terms from repeated or intersecting cycles. It therefore majorizes one plus the family sum in (26), which proves (30). If $\Xi_\phi < \log 2$, then $\mathcal{A}_\phi \leq e^{\Xi_\phi} - 1 < 1$; substituting this estimate into (27) gives (32). $\square$

Remark 4.6 (The $\log 2$ threshold is sufficient, not sharp). The sharp phase-blind threshold applies to $\mathcal{A}_\phi$, not to its exponential majorant. In the two-cycle model (28), put

$$xy = q \in (\log 2, 1).$$

There is only one simple cycle, so

$$\Xi_\phi = \mathcal{A}_\phi = q, \quad \det(I + E) = 1 - q > 0.$$

Thus the exact activity certificate succeeds although $\Xi_\phi < \log 2$ fails. The loss comes solely from allowing arbitrary ordered tuples of repeated and intersecting cycles in the exponential.

4.3 A degree–entry-size bound

Put

$$\begin{aligned} q_\phi &:= |\{(i, j) : i \neq j, (E_\phi)_{i,j} \neq 0\}|, \\ \Delta_\phi^+ &:= \max_{i \in V} |\{j \neq i : (E_\phi)_{i,j} \neq 0\}|, \\ \eta_\phi &:= \max_{i \neq j} |(E_\phi)_{i,j}|, \end{aligned} \quad (33)$$

with $\eta_\phi = 0$ when $q_\phi = 0$, and write $n = |V|$.

Proposition 4.7 (Cycle mass from out-degree and pivot ratios). *For every matching-normalized core,*

$$\boxed{\Xi_\phi \leq \sum_{k=2}^n \frac{q_\phi}{k} (\Delta_\phi^+)^{k-2} \eta_\phi^k.} \quad (34)$$

If $\Delta_\phi^+ \eta_\phi < 1$, then

$$\Xi_\phi \leq \frac{q_\phi \eta_\phi^2}{2(1 - \Delta_\phi^+ \eta_\phi)} \quad (35)$$

$$\leq \frac{n \Delta_\phi^+ \eta_\phi^2}{2(1 - \Delta_\phi^+ \eta_\phi)}. \quad (36)$$

Consequently,

$$\mathcal{A}_\phi \leq \exp \left\{ \frac{q_\phi \eta_\phi^2}{2(1 - \Delta_\phi^+ \eta_\phi)} \right\} - 1, \quad (37)$$

and the explicit sufficient condition

$$\frac{q_\phi \eta_\phi^2}{2(1 - \Delta_\phi^+ \eta_\phi)} < \log 2 \quad (38)$$

implies the positive determinant margin (32).

Proof. For a directed simple cycle of length k , choose one of its k oriented edges as the starting edge. There are at most q_ϕ choices for that edge and at most $(\Delta_\phi^+)^{k-2}$ choices for the next $k-2$ edges; the closing edge, if present, is then fixed. Since each cycle was counted k times, the number of length- k cycles is at most

$$\frac{q_\phi}{k} (\Delta_\phi^+)^{k-2}.$$

Every such cycle has absolute weight at most η_ϕ^k , proving (34).

When $\Delta_\phi^+ \eta_\phi < 1$, use $1/k \leq 1/2$ for $k \geq 2$ and sum the resulting geometric series. This gives (35). The inequality $q_\phi \leq n \Delta_\phi^+$ gives (36). Finally apply [theorem 4.5](#). $\square$

Remark 4.8 (What the combinatorial bound does not supply). The entry parameter is the pivot ratio

$$\eta_\phi = \max_{i \neq j} \frac{|t_{\phi(i),j}|}{|t_{\phi(j),j}|},$$

not merely the largest unnormalized collision coefficient. Moreover, TPC-66's reduced pair-codegree estimate does not by itself bound q_ϕ , Δ_ϕ^+ , or η_ϕ for an actual full (γ, s, j) -minor [2]. The present activity estimates are therefore L0 certificates. They become L1 only after the literal minor, matching, pivot ratios, and full decorated collision core have been identified. No polynomial physical determinant bound, parity improvement, or prime-pair conclusion is asserted here.

5 Determinant mass and the sharp lower-spectrum conversion

The preceding interface fixes the literal normalized full-output matrix B_X , its unit-diagonal Gram

$$K_X = B_X^* B_X,$$

the number $r_X \geq 1$ of active physical rows, and the full determinant mass

$$\mathfrak{D}_X = \sum_{\substack{I \subseteq \mathcal{Y}_X^{\text{full}} \\ |I|=r_X}} |\det B_X[I]|^2.$$

No output quotient or selected-row restriction is made in this section.

5.1 The exact Cauchy–Binet identity

Theorem 5.1 (Full determinant mass is the Gram determinant). *For every $r_X \geq 1$,*

$$\boxed{\mathfrak{D}_X = \det(B_X^* B_X) = \det K_X.} \quad (39)$$

The identity also holds when $|\mathcal{Y}_X^{\text{full}}| < r_X$: both sides then vanish.

Proof. Fix the orderings from the full-minor interface. Cauchy–Binet applied to $B_X^* B_X$ gives

$$\det(B_X^* B_X) = \sum_{\substack{I \subseteq \mathcal{Y}_X^{\text{full}} \\ |I|=r_X}} \det(B_X[I]^*) \det B_X[I].$$

Each summand is $|\det B_X[I]|^2$, which proves (39). If the output space has dimension less than r_X , then B_X has rank less than r_X , while the minor sum is empty by convention; hence both sides are zero. $\square$

For later use, define

$$c_1 := 1, \quad c_r := \left(\frac{r-1}{r} \right)^{r-1} \quad (r \geq 2). \quad (40)$$

Theorem 5.2 (Sharp determinant-to-gap conversion). *Let $K \succeq 0$ be an $r \times r$ matrix with unit diagonal, and put $\beta = \lambda_{\min}(K)$. If $r = 1$, then*

$$\beta = \det K = 1.$$

If $r \geq 2$, then

$$\det K \leq \beta \left(\frac{r-\beta}{r-1} \right)^{r-1} \leq \frac{\beta}{c_r}. \quad (41)$$

Consequently, for the literal matrix,

$$\boxed{\beta_X \geq c_{r_X} \mathfrak{D}_X \geq e^{-1} \mathfrak{D}_X.} \quad (42)$$

For every fixed $r \geq 2$, c_r is the largest universal constant for which

$$\lambda_{\min}(K) \geq c_r \det K$$

holds for every unit-diagonal positive semidefinite K .

Proof. The case $r = 1$ is immediate. Suppose $r \geq 2$, and list the eigenvalues as

$$0 \leq \lambda_1 = \beta \leq \lambda_2 \leq \dots \leq \lambda_r.$$

The unit diagonal gives

$$\sum_{j=1}^r \lambda_j = \text{tr } K = r.$$

The arithmetic–geometric mean inequality applied to $\lambda_2, \dots, \lambda_r$ yields

$$\prod_{j=2}^r \lambda_j \leq \left(\frac{r-\beta}{r-1} \right)^{r-1} \leq \left(\frac{r}{r-1} \right)^{r-1}.$$

Multiplication by β proves (41). Combine it with [theorem 5.1](#) to obtain the first inequality in (42). Moreover,

$$c_r = \left(1 + \frac{1}{r-1} \right)^{-(r-1)} \geq e^{-1},$$

which proves the second.

It remains to prove sharpness. For $0 < \varepsilon \leq 1$, let

$$K_{r,\varepsilon} := \frac{r-\varepsilon}{r-1}I_r - \frac{1-\varepsilon}{r-1}\mathbf{1}\mathbf{1}^*. \quad (43)$$

This is the negative equicorrelation matrix with diagonal one. Its eigenvalue on $\mathbf{1}$ is ε , and every vector orthogonal to $\mathbf{1}$ has eigenvalue $(r-\varepsilon)/(r-1)$. Thus it is positive definite and

$$\begin{aligned} \lambda_{\min}(K_{r,\varepsilon}) &= \varepsilon, \\ \det K_{r,\varepsilon} &= \varepsilon \left(\frac{r-\varepsilon}{r-1} \right)^{r-1}. \end{aligned}$$

Therefore

$$\frac{\lambda_{\min}(K_{r,\varepsilon})}{\det K_{r,\varepsilon}} = \left(\frac{r-1}{r-\varepsilon} \right)^{r-1} \rightarrow c_r \quad (\varepsilon \downarrow 0).$$

Any universal multiplier larger than c_r fails for sufficiently small ε . $\square$

Remark 5.3 (What sharpness does and does not say). The negative equicorrelation family proves sharpness among all unit-diagonal Gram matrices of a fixed dimension. It does not assert that the literal fixed- h_0 matrix has this form. Any improvement for the physical carrier must use additional actual-support or coefficient structure.

5.2 Summing certified full minors

Corollary 5.4 (Certified-minor mass). *Let*

$$\mathcal{J}_X^{\text{cert}} \subseteq \{I \subseteq \mathcal{Y}_X^{\text{full}} : |I| = r_X\},$$

and suppose that for each $I \in \mathcal{J}_X^{\text{cert}}$ an actual coefficient calculation supplies a number $\eta_X(I) \geq 0$ such that

$$|\det B_X[I]| \geq \eta_X(I). \quad (44)$$

Then

$$\boxed{\mathfrak{D}_X \geq \sum_{I \in \mathcal{J}_X^{\text{cert}}} \eta_X(I)^2, \quad \beta_X \geq c_{r_X} \sum_{I \in \mathcal{J}_X^{\text{cert}}} \eta_X(I)^2.} \quad (45)$$

In particular, if L_X distinct certified minors all satisfy $|\det B_X[I]| \geq \eta_X$, then

$$\beta_X \geq c_{r_X} L_X \eta_X^2. \quad (46)$$

Proof. Retain only the indicated nonnegative summands in (39), apply (44), and then use [theorem 5.2](#). $\square$

Remark 5.5 (Activity labels are not numerical certificates). The preceding forced-matching and cycle-activity analysis can be used to construct $\mathcal{J}_X^{\text{cert}}$. For example, if the minor $B_X[I]$, a matching ϕ , and its forced factors satisfy [theorem 4.3](#), then one valid choice is

$$\eta_X(I) := P_{\text{for}}(I)P_{\phi}(I)(1 - \mathcal{A}_{\phi}(I))_+. \quad (47)$$

However, a combinatorial activity label by itself is not an admissible $\eta_X(I)$. The corollary begins only after the literal determinant expansion has been shown to have the numerical lower bound (44). Conversely, minors not covered by the chosen certificate family remain in $\mathfrak{D}_X$; omitting them from the certified sum cannot be used as evidence that they vanish.

Remark 5.6 (Claim level). [Theorems 5.1](#) and [5.2](#) and [corollary 5.4](#) are finite-dimensional L0 statements. Their L1 specialization uses the already fixed literal full-output matrix B_X . They do not prove that a certified full minor exists, that the certified mass is polynomially large, or that $\mathfrak{D}_X$ avoids exponential decay as r_X grows. No parity or prime-pair conclusion follows from the determinant identity alone.

6 Exact decomposition by full-support collision components

The determinant mechanisms in the preceding sections can be tested one actual collision component at a time, but the components must then be reassembled with the correct quantifier. This is an exact orthogonal decomposition of the full matrix B_X , not permission to retain only a favorable component.

6.1 The full-support row collision graph

Define the row collision graph $\mathcal{G}_X^{\text{full}}$ on $\mathcal{M}_X^{\text{act}}$ by

$$m \sim n \iff m \neq n \text{ and there is } y \in \mathcal{Y}_X^{\text{full}} \text{ with } (B_X)_{y,m}(B_X)_{y,n} \neq 0. \quad (48)$$

Let $\mathcal{C}_X$ be its connected components. Isolated active rows are included as one-vertex components. For $C \in \mathcal{C}_X$, put

$$\mathcal{Y}_C := N_X^{\text{full}}(C) = \{y : (B_X)_{y,m} \neq 0 \text{ for some } m \in C\}, \quad (49)$$

and define the literal component matrix

$$B_C := B_X[\mathcal{Y}_C, C] : \ell^2(C) \longrightarrow \ell^2(\mathcal{Y}_C). \quad (50)$$

Lemma 6.1 (Disjoint output support of distinct components). *If $C, C' \in \mathcal{C}_X$ and $C \neq C'$, then*

$$\mathcal{Y}_C \cap \mathcal{Y}_{C'} = \emptyset. \quad (51)$$

Moreover,

$$\mathcal{Y}_X^{\text{full}} = \bigsqcup_{C \in \mathcal{C}_X} \mathcal{Y}_C. \quad (52)$$

Proof. If y belonged to both output sets, it would support nonzero entries from rows $m \in C$ and $n \in C'$. Then $m \sim n$ in $\mathcal{G}_X^{\text{full}}$, contradicting the assumption that the rows lie in different components. Every full output has a nonzero entry by (3), so it belongs to the output set of the component containing that row. $\square$

Proposition 6.2 (Exact block diagonalization). *After permutation of full-output and active-row coordinates,*

$$B_X = \bigoplus_{C \in \mathcal{C}_X} B_C, \quad (53)$$

$$K_X = B_X^* B_X = \bigoplus_{C \in \mathcal{C}_X} K_C, \quad K_C := B_C^* B_C. \quad (54)$$

Every column of every B_C still has norm one.

Proof. By lemma 6.1, a row in C has no nonzero matrix entry in $\mathcal{Y}_{C'}$ for $C' \neq C$. This proves the first orthogonal direct-sum identity. Taking the Gram gives the second. Component restriction removes no output touched by a column in C , so the unit-column identity (7) remains unchanged. $\square$

6.2 Componentwise lower-frame and kernel reduction

For each component put

$$\beta_C := \lambda_{\min}(K_C) = \sigma_{\min}(B_C)^2. \quad (55)$$

Theorem 6.3 (Exact componentwise lower-frame law). *For $x = \bigoplus_C x_C$,*

$$\|B_X x\|^2 = \sum_{C \in \mathcal{C}_X} \|B_C x_C\|^2. \quad (56)$$

Consequently,

$$\boxed{\beta_X = \min_{C \in \mathcal{C}_X} \beta_C}, \quad (57)$$

$$\ker B_X = \bigoplus_{C \in \mathcal{C}_X} \ker B_C, \quad (58)$$

$$\dim \ker B_X = \sum_{C \in \mathcal{C}_X} \dim \ker B_C. \quad (59)$$

In particular, B_X is injective if and only if every component matrix B_C is injective.

Proof. Equation (56) is the squared-norm identity for the orthogonal direct sum in (53). It gives the lower bound

$$\|B_X x\|^2 \geq \left(\min_C \beta_C \right) \sum_C \|x_C\|^2.$$

Conversely, a unit vector supported on a component attaining the minimum shows equality in (57). The kernel and nullity formulas follow from the same direct sum. $\square$

Thus a component with a very good lower frame cannot compensate for one singular component. An isolated active row has $\beta_C = 1$, but this does not raise the minimum if another component is bad.

6.3 Support and rank defects factor componentwise

For $C \in \mathcal{C}_X$, define the local support Hall defect and literal rank defect by

$$\mathfrak{h}_C^{\text{supp}} := \max_{S \subseteq C} (|S| - |N_X^{\text{full}}(S)|)_+, \quad (60)$$

$$\mathfrak{h}_C^{\text{act}} := \max_{S \subseteq C} (|S| - \text{rank } B_X[N_X^{\text{full}}(S), S]). \quad (61)$$

Proposition 6.4 (Componentwise defect accounting). *The two global defects satisfy*

$$\mathfrak{h}_X^{\text{supp}} = \sum_{C \in \mathcal{C}_X} \mathfrak{h}_C^{\text{supp}}, \quad (62)$$

$$\mathfrak{h}_X^{\text{act}} = \sum_{C \in \mathcal{C}_X} \mathfrak{h}_C^{\text{act}} = \sum_{C \in \mathcal{C}_X} \dim \ker B_C. \quad (63)$$

Hence ordinary full-support Hall holds globally if and only if it holds inside every component, and the literal full-rank condition from [theorem 2.3](#) holds globally if and only if it holds inside every component.

Proof. For $S \subseteq \mathcal{M}_X^{\text{act}}$, write $S_C = S \cap C$. The component output sets are disjoint, so

$$|S| - |N_X^{\text{full}}(S)| = \sum_C (|S_C| - |N_X^{\text{full}}(S_C)|).$$

Allowing $S_C = \emptyset$ gives zero contribution, so maximizing the positive deficiency independently in each component proves (62). The rank-defect identity follows either by the same block-rank calculation or from [theorems 2.3](#) and [6.3](#). $\square$

For $C \in \mathcal{C}_X$, let $r_C = |C|$ and define its literal full determinant mass by

$$\mathfrak{D}_C := \sum_{\substack{I_C \subseteq \mathcal{Y}_C \\ |I_C| = r_C}} |\det B_C[I_C]|^2. \quad (64)$$

Proposition 6.5 (Exact determinant-mass product). *The full determinant mass factors as*

$$\boxed{\mathfrak{D}_X = \prod_{C \in \mathcal{C}_X} \mathfrak{D}_C.} \quad (65)$$

Consequently $\mathfrak{D}_X > 0$ if and only if $\mathfrak{D}_C > 0$ for every component.

Proof. Consider a full row set $I \subseteq \mathcal{Y}_X^{\text{full}}$ with $|I| = r_X$, and put $I_C = I \cap \mathcal{Y}_C$. In the block matrix (53), its determinant can be nonzero only if

$$|I_C| = r_C \quad (C \in \mathcal{C}_X).$$

Indeed, if one block is assigned fewer than r_C output rows, that block cannot have full column rank; because the total numbers of selected rows and columns agree, an excess in another block cannot repair this deficit. When all equalities hold, permutation to component order gives

$$|\det B_X[I]|^2 = \prod_C |\det B_C[I_C]|^2.$$

Summing independently over the I_C 's proves (65). This is a block-support factorization; it does not reprove the preceding quantitative Cauchy–Binet lower inequality. $\square$

Corollary 6.6 (Componentwise determinant conversion). *With c_r as in (40),*

$$\boxed{\beta_X \geq \min_{C \in \mathcal{C}_X} c_{r_C} \mathfrak{D}_C \geq e^{-1} \min_{C \in \mathcal{C}_X} \mathfrak{D}_C.} \quad (66)$$

More generally, suppose that for each component C a family $\mathcal{J}_C^{\text{cert}}$ of literal full-column component minors has certified lower bounds

$$I_C \in \mathcal{J}_C^{\text{cert}}, \quad |\det B_C[I_C]| \geq \eta_C(I_C).$$

Then

$$\beta_X \geq \min_{C \in \mathcal{C}_X} c_{r_C} \sum_{I_C \in \mathcal{J}_C^{\text{cert}}} \eta_C(I_C)^2. \quad (67)$$

Proof. Every B_C has r_C unit columns by proposition 6.2. Cauchy–Binet gives $\mathfrak{D}_C = \det K_C$. Apply theorem 5.2 to each component Gram and combine the result with (57). Retaining only the certified nonnegative component-minor summands proves the last formula. $\square$

Remark 6.7 (Why the componentwise conversion matters). The global identity $\mathfrak{D}_X = \prod_C \mathfrak{D}_C$ can be exponentially smaller than every individual component mass when the number of components grows. Equation (66) pays no such gratuitous product: a uniform lower frame needs the worst component, not the product of all component determinants. This improvement is exact only for the full-support coordinate components above; it does not authorize an arbitrary row partition or deletion.

6.4 Full components are the mandatory scope

The reduced projection $(\gamma, s, j) \mapsto (s, j)$ can join two rows that occupy disjoint γ -coordinates. Therefore the reduced collision graph can merge several components of $\mathcal{G}_X^{\text{full}}$. It may be used as an arithmetic supergraph, but it is not the graph whose connected components produce the canonical coordinate block decomposition (53). In particular, reduced-cell codegree does not alter the identities (57) or (65).

Remark 6.8 (No favorable-component selection). The component partition is determined by the full actual nonzero support after all active rows have been included and before a test vector is chosen. Passing to components is lossless because (56) retains every summand. A uniform physical theorem must prove the desired estimate in every component: the global lower frame is the minimum of the local constants, the global kernel is their direct sum, and the determinant mass is their product. Deleting a singular or poorly conditioned component would change the coefficient space and is not an L1 theorem for the original physical map.

All results in this section are exact finite-dimensional L0 statements for B_X . Their L1 content is solely that B_X , its support graph, and its components are the literal fixed- h_0 , full-output objects from section 2. No componentwise positivity, polynomial determinant mass, parity gain, or prime-pair conclusion is asserted.

7 Sharp stop models and route distinctions

The determinant identity separates exact rank information from the strength of a particular quantitative certificate. The following models show that these stops must not be conflated.

7.1 Exponential determinant loss with a uniform spectral gap

Example 7.1 (The path Gram). *Let P_r be the path on r vertices, let $A(P_r)$ be its adjacency matrix, and fix*

$$0 < a < \frac{1}{2}.$$

Define the unit-diagonal matrix

$$K_r := I_r + aA(P_r). \tag{68}$$

Its eigenvalues are

$$1 + 2a \cos \frac{k\pi}{r+1}, \quad k = 1, \dots, r. \tag{69}$$

Consequently

$$\lambda_{\min}(K_r) = 1 - 2a \cos \frac{\pi}{r+1} > 1 - 2a > 0 \tag{70}$$

uniformly in r . In particular, K_r is a Gram matrix; one may take $B_r = K_r^{1/2}$.

Put $D_r := \det K_r$, with $D_0 := 1$. Expansion along an endpoint of the path gives

$$D_0 = D_1 = 1, \quad D_r = D_{r-1} - a^2 D_{r-2} \quad (r \geq 2). \tag{71}$$

If

$$q_{\pm} := \frac{1 \pm \sqrt{1 - 4a^2}}{2}, \tag{72}$$

then

$$D_r = \frac{q_+^{r+1} - q_-^{r+1}}{q_+ - q_-} \asymp_a q_+^r, \quad 0 < q_- < q_+ < 1. \tag{73}$$

Thus the exact determinant mass of the square synthesis $B_r = K_r^{1/2}$ decays exponentially, while its smallest Gram eigenvalue stays uniformly positive.

Proof. The path-adjacency spectrum is

$$\left\{ 2 \cos \frac{k\pi}{r+1} : 1 \leq k \leq r \right\},$$

which proves (69)–(70). The tridiagonal determinant expansion gives (71). Solving its characteristic equation $q^2 - q + a^2 = 0$ gives the displayed formula. Since $0 < q_-/q_+ < 1$, the factor

$$\frac{q_+(1 - (q_-/q_+)^{r+1})}{q_+ - q_-}$$

is bounded above and below by positive constants depending only on a , which proves the final comparison. $\square$

Remark 7.2 (A determinant-route stop is not a spectral stop). For the path family,

$$c_r \det K_r \asymp_a q_+^r$$

is exponentially small, although $\lambda_{\min}(K_r) > 1 - 2a$. Therefore failure to obtain a dimension-uniform lower bound through determinant mass does not prove that the actual lower frame is small. It only stops the determinant conversion as a sufficiently strong route; direct spectral-radius, Schur, forest, or other Gram estimates may still succeed.

7.2 Hall stops and activity-certificate stops

Example 7.3 (Support Hall deficiency is a genuine kernel). *Let*

$$B = \begin{pmatrix} 1 & 1 \end{pmatrix}.$$

Both unit columns have the same unique output neighbor. For the set of both columns, the support neighborhood has size one, so Hall deficiency is one. Moreover,

$$K = B^*B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \det K = 0, \quad \lambda_{\min}(K) = 0.$$

Here the support obstruction forces an actual kernel; this is a genuine spectral stop.

Example 7.4 (Positive cycle activity is only certificate failure). *For $|\omega| = 1$, let*

$$B_\omega := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & \omega \end{pmatrix}. \tag{74}$$

The support is the complete bipartite graph $K_{2,2}$. It satisfies all Hall inequalities, has two perfect matchings, and its square support contains one alternating four-cycle. Thus a unique-matching, forced-matching certificate is unavailable. Relative to either perfect matching, the nonidentity cycle term has magnitude one, so $\mathcal{A}_\phi = 1$ and the strict activity certificate $\mathcal{A}_\phi < 1$ is also unavailable.

Nevertheless,

$$\det B_\omega = \frac{\omega - 1}{2}. \tag{75}$$

For $\omega = 1$, matching terms cancel and the Gram has a kernel. For $\omega = -1$,

$$B_{-1}^* B_{-1} = I_2, \quad |\det B_{-1}|^2 = 1, \quad \lambda_{\min}(B_{-1}^* B_{-1}) = 1.$$

Hence activity at least one does not imply a kernel or even a small gap. It says only that support alone no longer certifies the literal determinant without controlling the alternating-cycle holonomy.

Remark 7.5 (Four distinct stop statements). The examples and the exact determinant identity give four logically different conclusions.

- (i) A support Hall deficiency forces $\mathfrak{D}_X = \beta_X = 0$. This is a support-level kernel stop.
- (ii) Hall may hold while literal matching terms cancel, as for B_1 . This is an actual coefficient kernel, not a support deficiency.
- (iii) Activity $\mathcal{A}_\phi \geq 1$, or failure of a unique-matching certificate, does not imply $\mathfrak{D}_X = 0$. The matrix B_{-1} has no such certificate but has determinant mass and spectral gap equal to one.
- (iv) Even an exactly known positive determinant mass may be exponentially small while the true spectral gap is uniform, as in [example 7.1](#). This stops only the determinant-to-gap route at the desired scale.

In particular, an empty certified-minor family gives a zero *certified lower sum*; it does not say that the omitted minors, the full determinant mass, or the lower frame vanish.

Remark 7.6 (Claim level). These are L0 finite-dimensional countermodels. They separate route diagnostics before any L1 arithmetic specialization. None asserts that the literal fixed- h_0 collision core realizes the path family, the two-matching phase, or a Hall defect. Establishing any such property for the physical coefficients is a separate arithmetic task.

8 Conclusions, stop rules, and the next door

8.1 What is now exact

The determinant route has four exact layers.

- (1) Every square actual minor factors into the pivots exposed by iterative degree-one peeling and a nonpeelable residual core. A residual core without a perfect matching has zero determinant.
- (2) If the residual core admits a declared perfect matching, all possible cancellation is encoded by vertex-disjoint alternating-cycle families and their gauge-invariant holonomies.
- (3) Dominant-matching activity gives a literal, phase-safe determinant certificate. It makes no randomness assumption.
- (4) Cauchy–Binet aggregates every full minor, and unit diagonal converts that determinant mass to the Gram gap with the sharp fixed-dimension factor c_r ; the bound $c_r \geq e^{-1}$ is uniform in the dimension.

Consequently, if the literal physical matrix satisfies

$$\mathfrak{D}_X := \sum_{\substack{Y_0 \subseteq \mathcal{Y}_X^{\text{full}} \\ |Y_0| = |\mathcal{M}_X^{\text{act}}|}} |\det(B_X)_{Y_0, \mathcal{M}_X^{\text{act}}}|^2 \geq X^{-\lambda_{\text{det}} + o(1)},$$

then its represented Gram has the same fixed-power exponent:

$$\beta_X \geq e^{-1} X^{-\lambda_{\text{det}} + o(1)}.$$

The determinant-to-gap conversion is L0; identifying B_X with the literal fixed- h_0 carrier is L1. Proving the displayed physical lower bound would be L2, and remains open. The sharper componentwise version requires only

$$\min_{C \in \mathcal{C}_X} \mathfrak{D}_C \geq X^{-\lambda_{\text{det}} + o(1)},$$

uniformly over all actual components, and gives the same conclusion by [corollary 6.6](#); it avoids the irrelevant product over mutually orthogonal collision components. This is likewise an unproved L2 premise.

8.2 What remains unavailable

TPC-66's reduced pair-codegree theorem does not supply:

- a full-output Hall matching involving every active row;
- lower bounds for matching pivots or their products;
- a bound below one for the activity of an actual core;
- control of long alternating cycles or γ -multiplicity;
- a polynomial lower bound for $\mathfrak{D}_X$, or, more usefully, a uniform lower bound for $\min_C \mathfrak{D}_C$ or for the corresponding worst-component certified-minor mass.

Deleting difficult rows, replacing full outputs by reduced cells, or selecting a generic coefficient filling would change the quantifier and is not allowed.

8.3 Stop rules

The following failures have different force.

- (i) A full-support Hall deficiency forces $\mathfrak{D}_X = 0$. More generally, $\mathfrak{D}_X = 0$ is equivalent to a genuine kernel for the represented map, even when the support satisfies Hall's condition.
- (ii) $\mathcal{A}_\phi \geq 1$ kills only the chosen dominant-matching certificate. The same minor may still be nonsingular.
- (iii) A small global product $\mathfrak{D}_X = \prod_C \mathfrak{D}_C$ alone is not a stop: many orthogonal components can make that product exponentially small while every component gap is good. Failure of one chosen certified-minor family gives a stop only for that family. The present determinant conversion is genuinely too weak at the desired fixed-power scale only if the *actual* worst-component mass $\min_C \mathfrak{D}_C$, or the actual mass of the single connected component under study, is proved to be $X^{-\omega(1)}$. Even then it does not show that the Gram gap is small. The connected path countermodel proves the latter distinction sharply.

8.4 Route update for TPC-68

The determinant door is now fully specified but still needs new physical matching, pivot, and cycle-activity input. The next independent door is the signed native/reassembly map, where the common residual Möbius gauge has already been removed and genuine two-affine signs first reappear. TPC-68 should isolate exact pair, star, and operator cancellation certificates there, test them against the actual full-output normalization, and state a hard stop if only support or generic-phase information is available. No parity, fixed-2, prime-pair, or twin-prime conclusion follows from the present paper.

References

- [1] Liang Wang. Activated-range represented grams at a fixed shift: Repair invariance, multiblock kernel fusion, and the failure of row capacity transfer, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-65-activated-range-represented-gram. Preprint.

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